

HealthCare Management System (HCMS)

The HealthCare Management System aims to streamline and automate various aspects of any healthcare operations to enhance efficiency, accuracy, and patient experience. Each team will be assigned a unique specific (HCMS) to work on. The project will encompass various aspects of HCMS operations including patient management, appointment scheduling, medical record keeping, billing, and reporting.

1. System Requirements:

The system should encompass the following functionalities:

a) User Authentication and Authorization:

- Secure login for administrators, clinic staff, and patients.
- Role-based access control (Admin, Clinic Staff, Patient).

b) Patient Portal:

- Patient registration and profile management.
- Medical history and record access.
- Prescription and treatment plan management.
- Billing and payment processing.

c) Appointment Management:

- Schedule and manage appointments efficiently.

2. Technology Stack:

a) **Programming Languages:** Specify preferred languages (e.g., JavaScript, Python, etc.).

b) **Frameworks and Libraries:** Choose suitable frameworks for frontend and backend development (e.g., React, Node.js, Django, etc.).

c) **Database:** Select an appropriate database management system (e.g., PostgreSQL, MongoDB, etc.) and design the database schema.

3. Project Timeline

Phases	Deliverables	Deadline
Phase 0 (by Email)	• Chosen idea • Used Tools & language (All of them: frontend, backend, and so on). • Team Positions and your System Name.	Week 7
Phase 1 (Video+ Report)	• Task division • System Design • ER Diagram, and features choosing and assigning should be delivered in this phase.	Week 10
Phase 2 - Final Phase Presentation and Video Demo (20 minutes)	• 100% of the project should be finished. • All the other deliverables should be ready. • Deliver a presentation with a working prototype of your project.	Week 13